《通信与网络》
Communications & Networks

# Switching & Routing

## Xiaofeng Zhong

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# All of we have pre-conception


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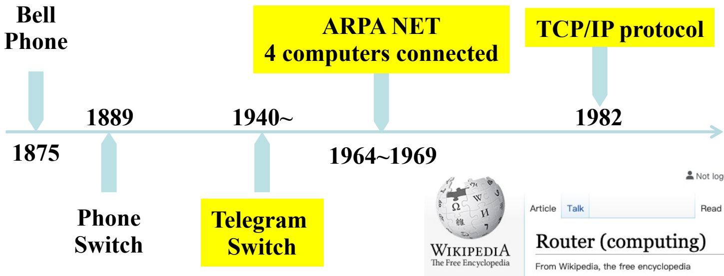
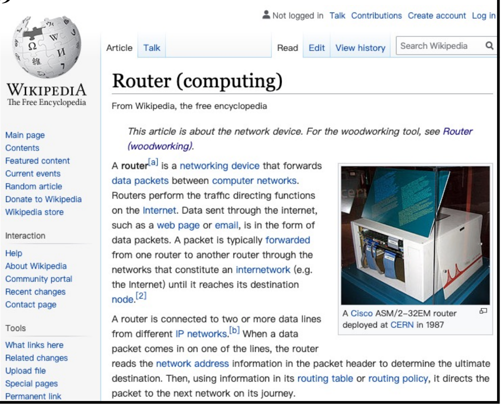
# History



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Router (computing)

From Wikipedia, the free encyclopedia

*This article is about the network device. For the woodworking tool, see Router (woodworking).*

A **router**[a] is a networking device that forwards data packets between computer networks. Routers perform the traffic directing functions on the Internet. Data sent through the internet, such as a web page or email, is in the form of data packets. A packet is typically forwarded from one router to another router through the networks that constitute an internetwork (e.g. the Internet) until it reaches its destination node.[2]

A router is connected to two or more data lines from different IP networks.[b] When a data packet comes in on one of the lines, the router reads the network address information in the packet header to determine the ultimate destination. Then, using information in its routing table or routing policy, it directs the packet to the next network on its journey.

A Cisco ASM/2-32EM router deployed at CERN in 1987

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# Addressing (at a conceptual level)

- Assume all hosts have unique IDs
- No particular structure to those IDs
- Later in course will talk about real IP addressing

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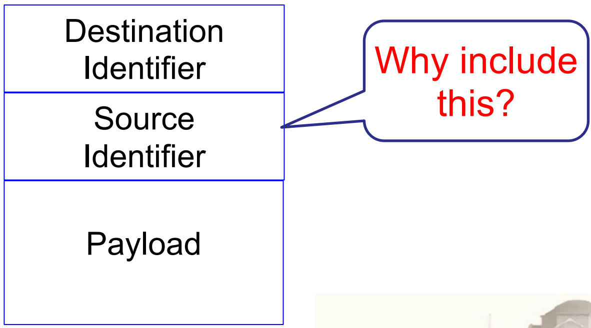
## Packets (at a conceptual level)

- **Assume packet headers contain:**
  - ❑ Source ID, Destination ID, and perhaps other information

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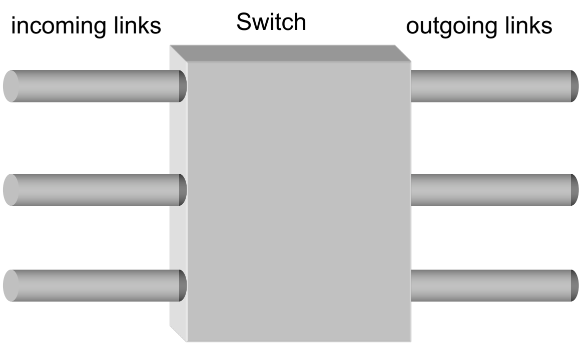
## Switches

- **Multiple ports (attached to other switches or hosts)**




- **Ports are typically duplex (incoming and outgoing)**

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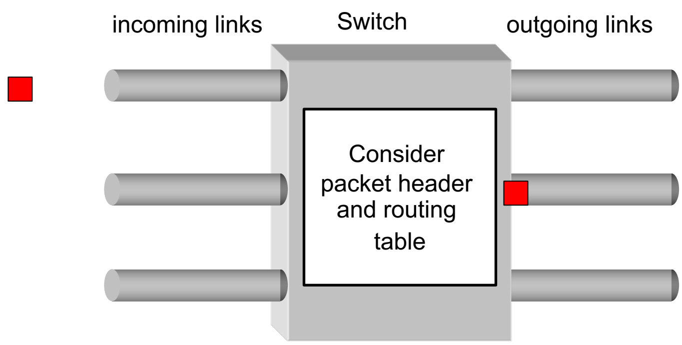
## Forwarding Decisions
## （Router is coming ）

- When packet arrives, must choose outgoing port




- Decision is based on routing state (table) in switch

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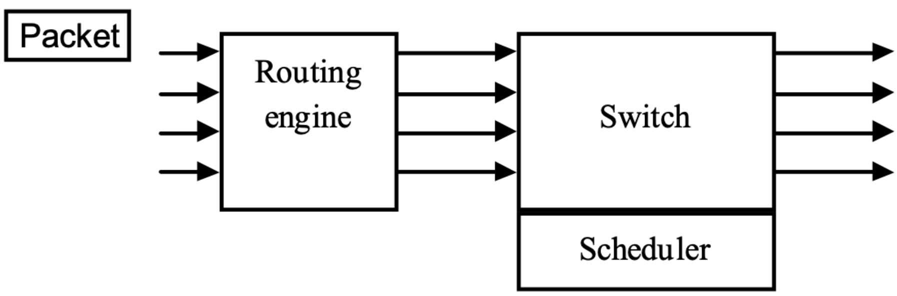
## Forwarding Decisions （Router）

- When packet arrives, must choose outgoing port
- Decision is based on routing state (table) in switch

Router is under background, while switch is the actor.

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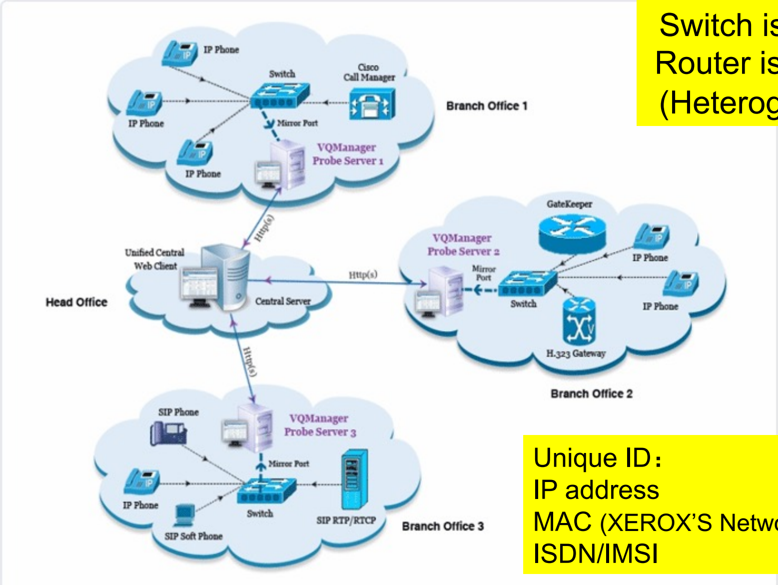
# Switch vs. Router




Switch is intra-net.
Router is inter-nets
(Heterogeneous）

Unique ID：
IP address
MAC (XEROX'S Network System)
ISDN/IMSI

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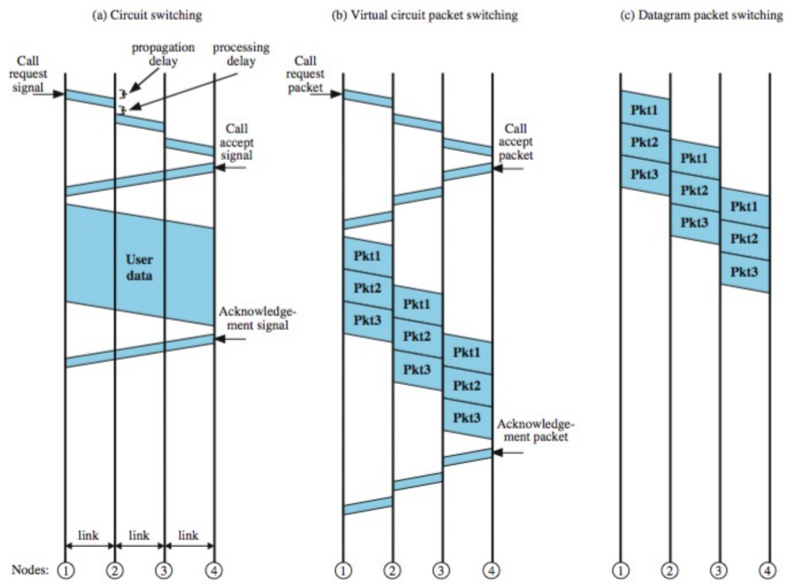
# Switch on Tele Line Network

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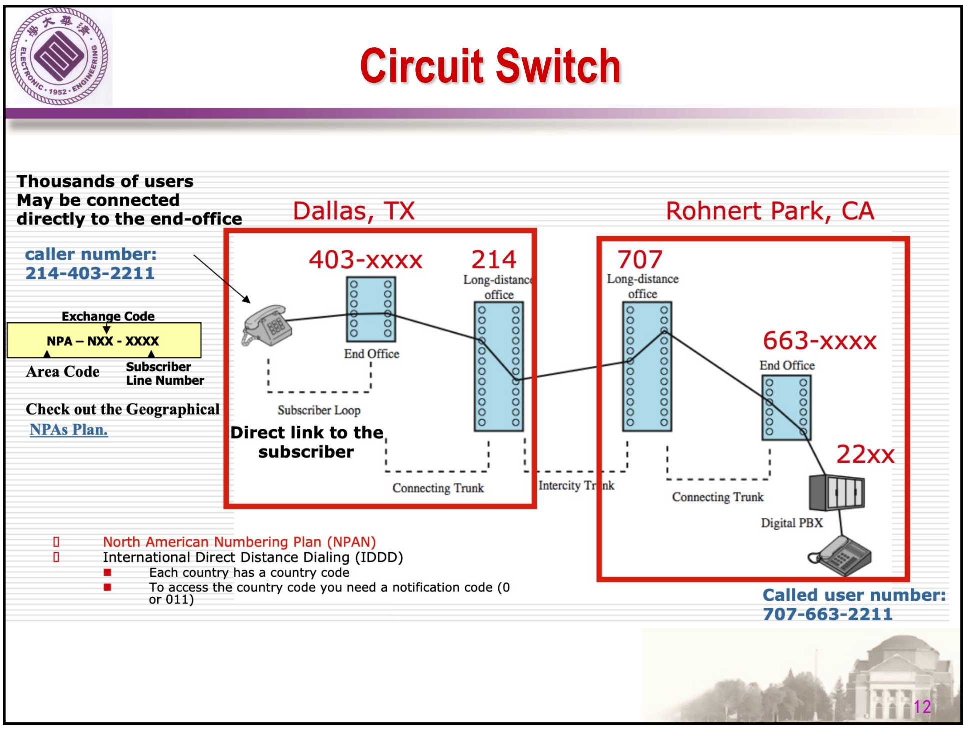
Circuit Switch
Thousands of users
May be connected
directly to the end-office
caller number:
214-403-2211
Exchange Code
NPA – NXX - XXXX
Area Code
Subscriber
Line Number
Check out the Geographical
NPAs Plan.
Dallas, TX
403-xxxx
214
Long-distance
office
End Office
Subscriber Loop
Direct link to the
subscriber
Connecting Trunk
Intercity Trunk
Rohnert Park, CA
707
Long-distance
office
663-xxxx
End Office
22xx
Connecting Trunk
Digital PBX
North American Numbering Plan (NPAN)
International Direct Distance Dialing (IDDD)
Each country has a country code
To access the country code you need a notification code (0 or 011)
Called user number:
707-663-2211
12

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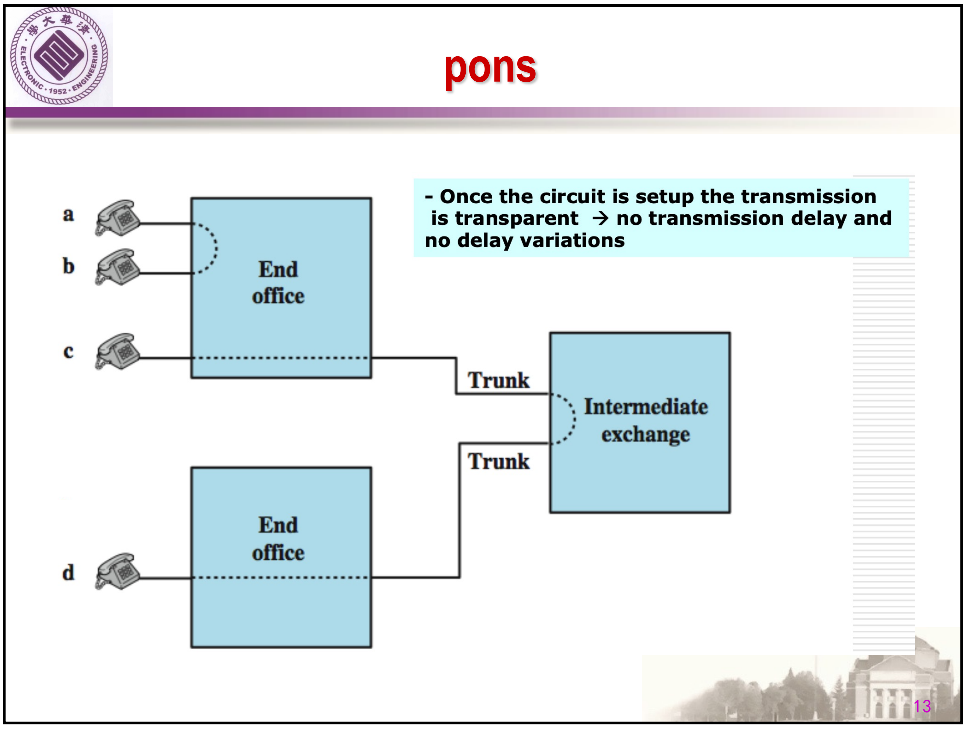
pons
- Once the circuit is setup the transmission is transparent → no transmission delay and no delay variations
a
b
c
d
End
office
Trunk
Intermediate
exchange
Trunk
End
office
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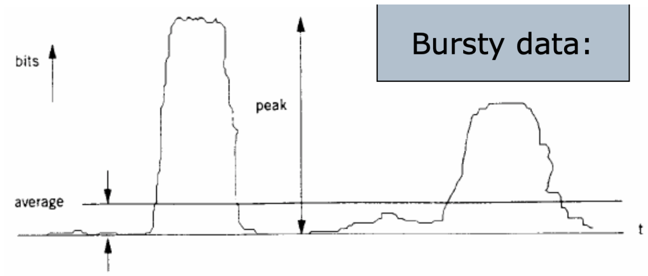
## cons

- set-up time is 150 msec， data length is 1000 bytes transmitting at 64Kbps.
  - Only takes 125 msec to transmit the data!
- Not efficient when data occurs in bursts separated by idle periods

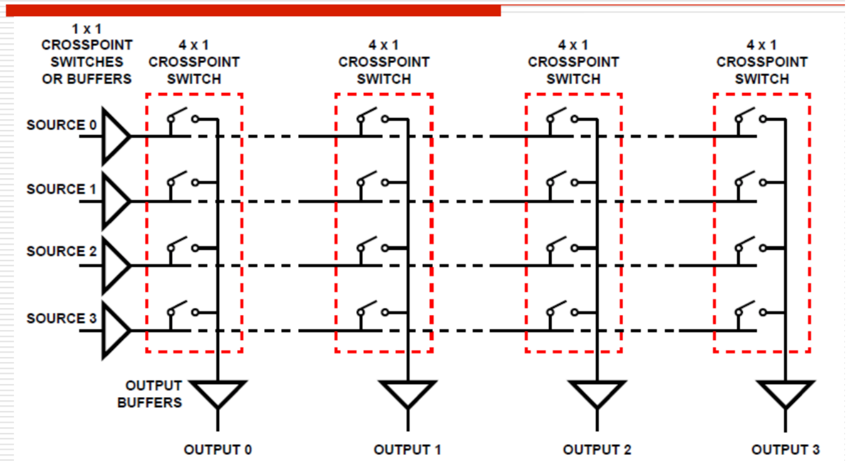
Bursty data:








## Switch Fabric using Crosspoints




Space Division Switching

This is a non-blocking Switch

## non-blocking network

- permits all stations to connect at once
- used for some data connections

## blocking network

- may be unable to connect stations because all paths are in use
- used on voice systems (users do not have calls all the time)

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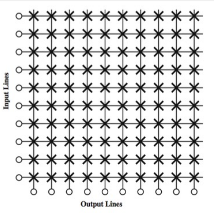
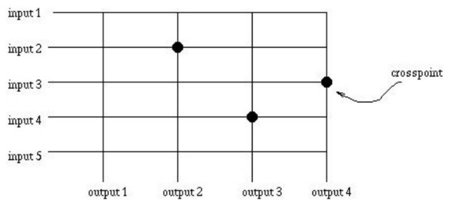
# Space Division Switch - Non-blocking

- NxN crosspoints
- For 10^6 users, a crossbar would require 10^12 crosspoints (N^2)




Input 2 is connected to ouput 2, input 3 is connected to output 4, input 4 is connected to output 3.

Note: Connections between free inputs and outputs are always possible.

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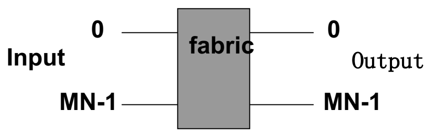
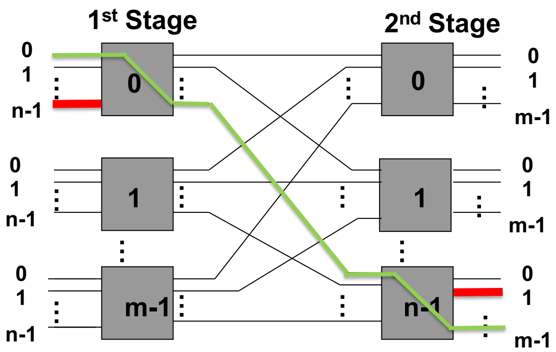
# Multiple stage Switch




Cross points: $MN * MN$




Cross points: $MN *$ （$N + M$）

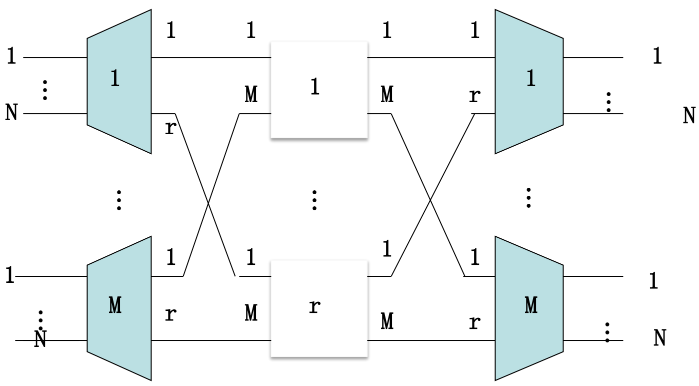
**Blocked Switch (Output)**

Nonblocking Switch?





# 3 Stage Space Division Switch —Clos Network



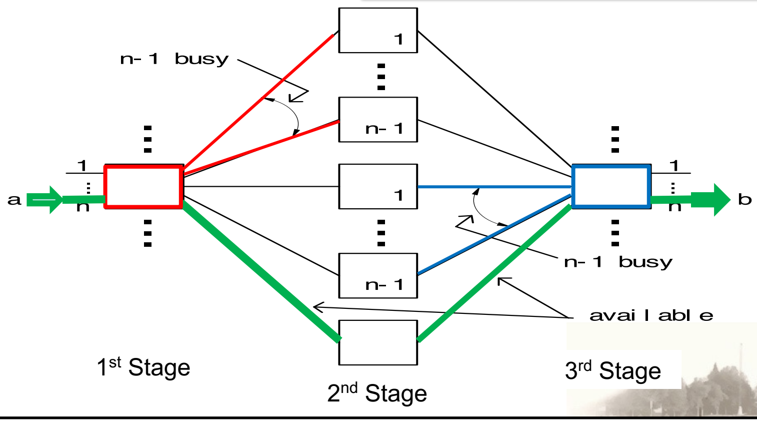

Clos, Charles (Mar 1953). "A study of non-blocking switching networks". Bell System Technical Journal 32 (2): 406 – 424.1538-7305.1953

# 3 stage, Clos Switch Network

**Worst case: (N-1)x2 fabric are occurred.**

**Non-blocking: $r \geq (N-1)X2+1=2N-1$**

**Re-construction non-blocking: $r \geq N$**





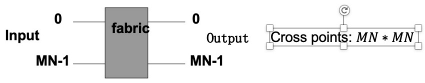

# 3 Stage, Clos Switch Network

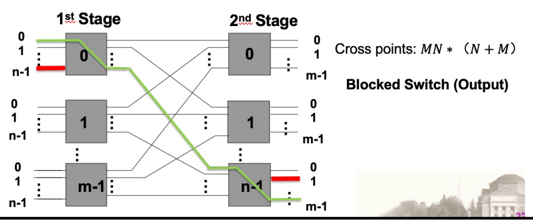
- **Size (cross points): $C_3 = 2M \times N \times r + r \times M^2$**
- **CLOS non-blocking structure: r=2N-1**

**$C_3 = M(2N+M)(2N-1)$**

**while N=M**

**$C_3 = 6N^3 - 3N^2 = O(N^3) < O(N^4)$**

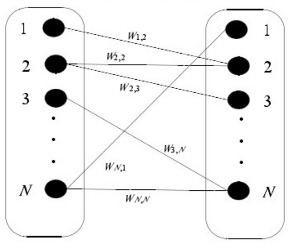


# Time Division Switch

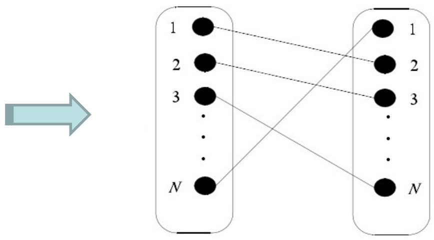
- Mostly all modern circuit switches are time-division switches.
- Time-slot interexchange (TSI)
- There is no need for address bits in each slot (synchronous)
- The slot could be a bit, a byte or a longer block.

Bipartite graph $G$




(a)

Matching $M$




(b)

Matching Problem

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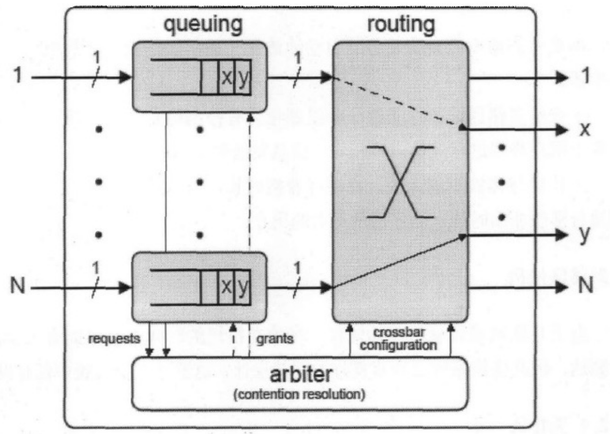
# Input Queuing




Output Blocking

In each cycle: 1 write, 1 read
Queuing cell speed = 2* port speed

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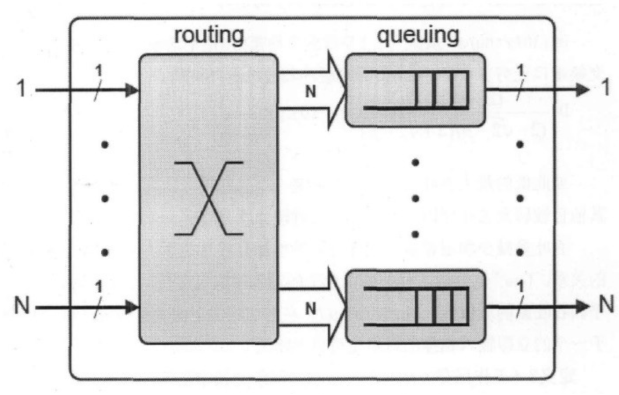
# Output Queuing




Input Blocking

In each cycle: N write, 1 read
Queuing cell speed = (N+1) * port speed



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